Chapter 20: Treatment-Confounder Feedback

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Chapter 19 identified sequential exchangeability as a key condition for identifying the causal effects of time-varying treatments. Suppose a study satisfies the strongest form of sequential exchangeability, so the measured time-varying confounders suffice to estimate the effect of any treatment strategy. Which confounding adjustment method should we then use? The answer exposes a central problem for time-varying treatments: a time-varying confounder can drive later treatment while itself being affected by earlier treatment, or while sharing with earlier treatment a cause outside that treatment's measured past, through a path that conditioning on that measured past does not block. When that happens, traditional adjustment methods can be biased even though all the information needed for valid estimation is available. This chapter describes the structure of the problem and explains why traditional methods fail.

This chapter is based on Hernán and Robins (2020, chap. 20, pp. 267-275).

Key insight: when a time-varying confounder L_1 that affects later treatment A_1 is also affected by earlier treatment A_0, or shares with A_0 a cause outside the measured past of A_0 through a path that conditioning on that measured past does not block, stratifying on L_1 removes confounding for the later treatment A_1 but opens a collider path that creates selection bias for A_0. Not adjusting for L_1 leaves A_1 confounded. Either way the estimate of a treatment-strategy effect is biased; g-methods (Chapter 21) avoid both problems.

1 20.1 The Elements of Treatment-Confounder Feedback (pp. 267-269)

Definition 1.1 (Treatment-Confounder Feedback)**.** Consider a time-varying treatment A_k and a time-varying confounder L_k, measured at times $k = 0, 1, \ldots$. For $j \geq 0$, write $(\bar{L}_j, \bar{A}_{j-1})$ for the measured past of treatment A_j. There is **treatment-confounder feedback** when, for some $k \geq 1$, the time-varying confounder affects treatment $(L_k \to A_k)$ *and*, for some earlier treatment A_j with $j < k$, either

- L_k is affected by A_j, or
- L_k and A_j share a common cause outside the measured past of A_j, through a path between them (such as $A_j \leftarrow W \to L_k$ with W unmeasured) that conditioning on that measured past does not block.

The book's one-line description is the effect form (a directed path $A_j \to \cdots \to L_k$ for some $j < k$): "the confounder affects the treatment and the treatment affects the confounder" (Hernán and Robins 2020, 267). The book gives the same name to the case in which L_1 shares an unmeasured cause W_0 with A_0 (its Figures 20.4 and 20.6), because conditioning on L_1 produces the same bias in that case as in the effect form; the observed data also cannot tell the two cases apart (Hernán and Robins 2020, 269 and 272). The second clause of Definition 1.1 covers that case only. It excludes a common cause in the measured past, such as an earlier confounder L_{k-1} that affects both A_{k-1} and L_k; otherwise almost every persistent confounder would count as feedback.

Example 1.1 (The Sequentially Randomized HIV Trial)**.** Return to the sequentially randomized HIV trial of Chapter 19:

- treatment A_k (1: treated, 0: untreated) at each month $k = 0, 1, \ldots, K$;
- covariates L_k (CD4 cell count) at each month;
- an outcome Y measuring health status at month $K + 1$.

The time-varying covariates L_k are time-varying confounders: CD4 count L_k affects treatment A_k. In addition, treatment A_{k-1} raises future CD4 count L_k. So the trial has treatment-confounder feedback (Definition 1.1).

The book's Figure 20.1 (the same diagram as Figure 19.2) shows the first two months of the trial. As in Chapter 19, the example has no censoring, so we can focus on confounding.

1.1 Time-Varying Confounding Without Feedback

Remark 1.1 (Time-Varying Confounding Without Feedback). Time-varying confounding can occur *without* treatment-confounder feedback (Definition 1.1). In the book's Figure 20.2, the arrows from treatment to later L and U variables are deleted: L_k still confounds the effect of A_k, but no earlier treatment A_j $(j < k)$ affects it. Nor does L_k share with any earlier A_j a common cause outside the measured past of A_j in the sense of Definition 1.1: the diagram is a sequentially randomized trial, so the unmeasured U variables have no arrows into treatment, and every path from a common cause to A_j passes through the measured confounders $\bar{L}_j$, which block it. There is time-varying confounding but no feedback.

i Fine Point 20.1: Representing Feedback Cycles with Acyclic Graphs

A feedback loop between treatment and confounder can be drawn on an acyclic graph by unrolling it in time: $A_{k-1} \to L_k \to A_k \to L_{k+1} \to \cdots$. This representation requires discrete time: treatment and covariates may change during each interval $[k, k + 1)$, without specifying when within the interval. In practice the intervals can be made as short as the data's granularity

requires (months in the HIV example, where patients see their doctors at most monthly), and time is usually recorded in discrete units anyway (Hernán and Robins 2020, Fine Point 20.1, p. 268).

1.2 The Simplest Case: Figure 20.3

Example 1.2 (The Simplest Feedback Diagram)**.** The smallest diagram that shows treatment-confounder feedback in a two-time-point sequentially randomized trial is the book's Figure 20.3. Its arrows are

$$A_0 \to L_1, \qquad L_1 \to A_1, \qquad U_1 \to L_1, \qquad U_1 \to Y.$$

It makes four simplifications relative to Figure 20.1:

- no baseline node L_0: A_0 is marginally randomized and A_1 is randomized conditional on L_1;
- no unmeasured U_0;
- no arrow $A_0 \to A_1$: treatment at time 1 is assigned using L_1 only;
- no arrows from A_0, L_1, or A_1 into Y: the sharp null[a] holds.

[a] 01-introduction.qmd#def-sharp-null

None of these simplifications affects the arguments of the chapter; a larger diagram would add no conceptual insight and would only be harder to read.

Remark 1.2 (The True Effect Under Figure 20.3 Is Zero). Because Figure 20.3 has no forward-directed path from A_0 or A_1 to Y, the true average causal effect of "always treat" versus "never treat" is zero:

$$\mathbb{E}\big[Y^{a_0=1,a_1=1}\big] - \mathbb{E}\big[Y^{a_0=0,a_1=0}\big] = 0.$$

There are no arrows from the unmeasured U_1 into the treatments, so Figure 20.3 can represent a sequentially randomized trial (or an observational study with no unmeasured confounding), and Chapter 19's results say the observed data on (A_0, L_1, A_1, Y) should identify this effect as 0.

1.3 Feedback Through Shared Causes: Figure 20.4

Example 1.3 (Feedback Through a Shared Cause)**.** The same problem arises when the time-varying confounder is not affected by prior treatment but **shares an unmeasured cause** W_0 with it (the book's Figure 20.4, a subset of Figure 19.4). Its arrows are

$$W_0 \to A_0, \qquad W_0 \to L_1, \qquad L_1 \to A_1, \qquad U_1 \to L_1, \qquad U_1 \to Y.$$

Figure 20.4 represents an observational study. The path $A_0 \leftarrow W_0 \to L_1$ runs through a cause outside the measured past of A_0, so this is the second clause of Definition 1.1. The book refers to both Figures 20.3 and 20.4 (and 19.2 and 19.4) as examples of treatment-confounder feedback (Definition 1.1).

Remark 1.3 (The Claim of This Chapter). With time-varying confounders and treatment-confounder feedback, traditional methods (stratification, matching, and outcome regression whose treatment coefficients are read as the effect) cannot correctly adjust for the confounders, even when the data suffice for sequential exchangeability. G-methods adjust correctly even in

the presence of feedback.

2 20.2 The Bias of Traditional Methods (pp. 269-271)

Example 2.1 (A Hypothetical Two-Time-Point Trial)**.** A hypothetical sequentially randomized trial with 32,000 individuals with HIV and two time points $k = 0, 1$ (full adherence, no loss to follow-up):

- $A_0 = 1$ is randomly assigned at baseline with probability 0.5;
- A_1 is randomly assigned at month 1 with probability 0.4 if $L_1 = 0$ (high CD4) and 0.8 if $L_1 = 1$ (low CD4);
- the outcome Y, measured at the end of follow-up, is a health score (higher is better).

The data are in Table 1.

Table 1: Each row is a combination of (A_0, L_1, A_1), with its number of individuals N and $\mathbb{E}[Y \mid A_0, L_1, A_1]$. Source: Hernán and Robins (2020, Table 20.1, p. 269).

N	A_0	L_1	A_1	Mean Y
2400	0	0	0	84
1600	0	0	1	84
2400	0	1	0	52
9600	0	1	1	52
4800	1	0	0	76
3200	1	0	1	76
1600	1	1	0	44
6400	1	1	1	44

Example 2.2 (Reading the Design Back From the Data)**.** The design probabilities can be read back from Table 1. Within $L_1 = 0$: $1600/(2400 + 1600) = 0.4$ and $3200/(4800 + 3200) = 0.4$. Within $L_1 = 1$: $9600/(2400 + 9600) = 0.8$ and $6400/(1600 + 6400) = 0.8$. Also, 16,000 of the 32,000 individuals have $A_0 = 1$, matching probability 0.5.

Remark 2.1 (Identifiability Holds by Design). The trial is built so that all three identifiability conditions hold (Hernán and Robins 2020, 269):

- **sequential exchangeability**: randomization leaves A_0 with no confounders, and A_1 is assigned using L_1 alone, so exchangeability for A_1 holds once we condition on L_1;
- **positivity**: each of the eight rows of the table contains individuals;
- **consistency**: each treatment history $\bar{a} = (a_0, a_1)$ is a well-defined intervention, and a person whose observed history is $\bar{a}$ has observed outcome $Y = Y^{\bar{a}}$; in this ideal trial the recorded history is the treatment actually received.

Had the trial run past month 1, with each month's treatment assigned using that month's CD4 count, each of those later CD4 measurements would be a time-varying confounder as well. Random variability is ignored throughout.

2.1 Each Treatment, Taken Separately, Has No Effect

Example 2.3 (No Effect of A_1)**.** In every stratum of (A_0, L_1), the mean outcome is the same for $A_1 = 1$ and $A_1 = 0$ (rows 1 vs 2, 3 vs 4, 5 vs 6, 7 vs 8). For example,

$$\mathbb{E}[Y \mid A_0 = 0, L_1 = 0, A_1 = 1] - \mathbb{E}[Y \mid A_0 = 0, L_1 = 0, A_1 = 0] = 84 - 84 = 0,$$

and because the identifiability conditions hold this equals $\mathbb{E}[Y^{a_1=1} \mid A_0 = 0, L_1 = 0] - \mathbb{E}[Y^{a_1=0} \mid A_0 = 0, L_1 = 0]$.

Example 2.4 (No Effect of A_0)**.** Because A_0 is randomized (so exchangeability holds for A_0 without adjustment), and positivity and consistency also hold (Remark 2.1), $\mathbb{E}[Y \mid A_0 = 1] - \mathbb{E}[Y \mid A_0 = 0]$ estimates $\mathbb{E}[Y^{a_0=1}] - \mathbb{E}[Y^{a_0=0}]$. Averaging rows 1-4 (16,000 individuals with $A_0 = 0$):

$$\begin{aligned}\mathbb{E}[Y \mid A_0 = 0] &= \frac{2400 \times 84 + 1600 \times 84 + 2400 \times 52 + 9600 \times 52}{16000} \\ &= \frac{4000 \times 84 + 12000 \times 52}{16000} = \frac{336000 + 624000}{16000} = 60.\end{aligned}$$

Averaging rows 5-8 (16,000 individuals with $A_0 = 1$):

$$\begin{aligned}\mathbb{E}[Y \mid A_0 = 1] &= \frac{4800 \times 76 + 3200 \times 76 + 1600 \times 44 + 6400 \times 44}{16000} \\ &= \frac{8000 \times 76 + 8000 \times 44}{16000} = \frac{608000 + 352000}{16000} = 60.\end{aligned}$$

So the average causal effect of A_0 is $60 - 60 = 0$.

ℹ Technical Point 20.1: G-Null Test

The **g-null theorem** (Robins 1986) ties the sharp null to two conditional independencies in the observed data. With two time points, the independencies are

$$Y \perp\!\!\!\perp A_0 \mid L_0 \quad \text{and} \quad Y \perp\!\!\!\perp A_1 \mid A_0, L_0, L_1.$$

Assume sequential randomization for every strategy g. The theorem says that both independencies hold exactly when the distribution of Y^g does not depend on g and coincides with the distribution of the observed Y, so that in particular $\mathbb{E}[Y^g] = \mathbb{E}[Y]$ for every g (Hernán and Robins 2020, Technical Point 20.1, p. 270).

To see why the sharp null leads to the independencies, note that if no strategy changes anyone's outcome, then $Y^g = Y$ for every g. Sequential exchangeability for the counterfactuals Y^g then becomes a statement about Y itself. That step relies on every pair (a_0, l_0) being produced by some strategy with $a_0 = g(l_0)$.

Each independency has a causal reading:

- the first says A_0 has no effect within any stratum of L_0;
- the second says A_1 has no effect within any stratum of (A_0, L_0, L_1), which in Figure 20.3 (where there is no L_0) means within strata of (A_0, L_1).

When sequential exchangeability holds, checking these independencies in the data therefore tests the sharp null. That check is the **g-null test** (Robins 1986).

2.2 Analyzing the Joint Treatment

The effects of A_0 and A_1 are each null. Each null taken alone does not establish the joint null for the strategy (a_0, a_1). The g-null theorem of Technical Point 20.1 does not apply directly, because

its premise is conditional independence and the examples check only conditional means. The mean joint null says that $\mathbb{E}[Y^{a_0,a_1}]$ is the same for every static strategy. The equal conditional means of Example 2.3 and Example 2.4 are enough for it, as computing the mean outcome under each static strategy with the g-formula shows. That computation uses all three identifiability conditions (Remark 2.1): sequential exchangeability (unconditionally for the randomized A_0, and given (A_0, L_1) for A_1), positivity, and consistency. Under these conditions,

$$\begin{aligned}
\mathbb{E}[Y^{a_0,a_1}] &= \mathbb{E}[Y^{a_0,a_1} \mid A_0 = a_0] && (A_0 \text{ randomized}) \\
&= \sum_l \mathbb{E}[Y^{a_0,a_1} \mid A_0 = a_0, L_1 = l] \Pr[L_1 = l \mid A_0 = a_0] && (\text{law of total expectation}) \\
&= \sum_l \mathbb{E}[Y^{a_0,a_1} \mid A_0 = a_0, L_1 = l, A_1 = a_1] \Pr[L_1 = l \mid A_0 = a_0] && (\text{sequential exchangeability}) \\
&= \sum_l \mathbb{E}[Y \mid A_0 = a_0, L_1 = l, A_1 = a_1] \Pr[L_1 = l \mid A_0 = a_0] && (\text{consistency}).
\end{aligned}$$

From Table 1, $\Pr[L_1 = 0 \mid A_0 = 0] = 4000/16000 = 0.25$ and $\Pr[L_1 = 0 \mid A_0 = 1] = 8000/16000 = 0.5$, and the cell means do not depend on a_1, so for either value of a_1

$$\begin{aligned}
\mathbb{E}\left[Y^{a_0=0,a_1}\right] &= 84 \times 0.25 + 52 \times 0.75 = 21 + 39 = 60, \\
\mathbb{E}\left[Y^{a_0=1,a_1}\right] &= 76 \times 0.5 + 44 \times 0.5 = 38 + 22 = 60.
\end{aligned}$$

So $\mathbb{E}[Y^{a_0,a_1}] = 60$ for every (a_0, a_1), and in particular $\mathbb{E}\left[Y^{a_0=1,a_1=1}\right] - \mathbb{E}\left[Y^{a_0=0,a_1=0}\right] = 0$. Do conventional analyses recover this?

Example 2.5 (Analysis 1: No Adjustment for L_1)**.** Compare the 9600 individuals treated at both times (rows 6 and 8) with the 4800 untreated at both times (rows 1 and 3):

$$\begin{aligned}
\mathbb{E}[Y \mid A_0 = 1, A_1 = 1] &= \frac{3200 \times 76 + 6400 \times 44}{9600} = \frac{243200 + 281600}{9600} \approx 54.7, \\
\mathbb{E}[Y \mid A_0 = 0, A_1 = 0] &= \frac{2400 \times 84 + 2400 \times 52}{4800} = \frac{201600 + 124800}{4800} = 68.0.
\end{aligned}$$

The difference $54.7 - 68 = -13.3$ is non-null, because $\mathbb{E}[Y \mid A_0 = a_0, A_1 = a_1]$ is not a valid estimator of $\mathbb{E}[Y^{a_0,a_1}]$: adjustment for the confounder L_1 is needed.

Example 2.6 (Analysis 2: Stratification on L_1)**.** Within $L_1 = 0$, compare treated at both times (row 6) with untreated at both times (row 1):

$$\mathbb{E}[Y \mid A_0 = 1, L_1 = 0, A_1 = 1] - \mathbb{E}[Y \mid A_0 = 0, L_1 = 0, A_1 = 0] = 76 - 84 = -8.$$

Within $L_1 = 1$ (rows 8 and 3): $44 - 52 = -8$.
The analysis adjusted for the confounder also gives the wrong answer.

Remark 2.2 (No Weighting of the Strata Recovers the Null). Because the stratum-specific difference is -8 in *both* strata of L_1, no weighted average of the stratum-specific differences can equal the correct value 0. This estimate reflects the bias of traditional methods when there is treatment-confounder feedback (Definition 1.1) (Hernán and Robins 2020, 271).

Example 2.7 (Preview of Chapter 21: The G-Formula)**.** The computation that gave $\mathbb{E}[Y^{a_0,a_1}] = 60$ for every strategy is the g-formula: it standardizes over the distribution of L_1 *given* A_0 rather than conditioning on L_1, as the stratified analysis does. Applied to "always treat" versus "never treat", it gives $60 - 60 = 0$, as it should. Chapter 21 develops this computation.

3 20.3 Why Traditional Methods Fail (pp. 271-273)

Remark 3.1 (The Problem Is the Adjustment Method). All three identifiability conditions hold in Table 1; the unmeasured U_1 (immunosuppression level) is not needed because we have data on L_1. Yet neither analysis gave the correct answer. The problem is the **adjustment method**: stratification cannot handle treatment-confounder feedback (Definition 1.1).

3.1 Stratification Opens a Collider Path

Remark 3.2 (Stratification Opens a Collider Path). Stratifying on L_1 means estimating the treatment-outcome association separately in the subsets $L_1 = 0$ (high CD4) and $L_1 = 1$ (low CD4). But L_1 is affected by prior treatment A_0, so it is a **collider** on the path

$$A_0 \to L_1 \leftarrow U_1 \to Y$$

(the book's Figure 20.5: Figure 20.3 with L_1 conditioned on). Conditioning on L_1 opens this path and generically (that is, under faithfulness) induces a noncausal association between A_0 and U_1, and hence between A_0 and Y, within levels of L_1.

Intuition: among those with low CD4 count ($L_1 = 1$), being on treatment ($A_0 = 1$) marks severe immunosuppression (high U_1), since treatment would otherwise have raised their CD4. Among those with high CD4 count ($L_1 = 0$), being off treatment ($A_0 = 0$) marks milder immunosuppression (low U_1).

Remark 3.3 (Trading Confounding for Selection Bias). Stratification eliminates confounding for A_1 at the cost of introducing selection bias for A_0. The associational differences

$$\mathbb{E}[Y \mid A_0 = 1, L_1 = l, A_1 = 1] - \mathbb{E}[Y \mid A_0 = 0, L_1 = l, A_1 = 0]$$

may be non-zero even when treatment has no effect on anyone's outcome at any time. The net bias depends on the relative sizes of the confounding removed and the selection bias created.

3.2 Feedback Through Shared Causes

Remark 3.4 (The Same Bias Under a Shared Cause). The same bias arises when the confounder shares an unmeasured cause W_0 with prior treatment (an observational study, the book's Figure 20.6): conditioning on the collider L_1 opens

$$A_0 \leftarrow W_0 \to L_1 \leftarrow U_1 \to Y.$$

Because conditioning on L_1 creates the same bias in Figure 20.4 as in Figure 20.3, both settings are called treatment-confounder feedback; the observed data also cannot tell them apart.

3.3 Beyond the Null and Beyond Two Time Points

The Bias Is Not Limited to the Null or to Two Time Points

- The bias is not limited to the sharp null. With a non-null effect (the book's Figure 20.7, where $A_0 \to L_1 \to Y$), adding arrows into Y from A_0, A_1, or L_1 leaves the collider path through L_1 in place, so conditioning on L_1 still generically (that is, under faithfulness) associates A_0 with Y in a way that has no causal interpretation.
- With many time points and variables, the bias still arises; confounders affected by prior treatment at multiple times increase the possibility of a large bias.
- Valid estimation of a treatment-strategy effect requires estimating the joint effect of all treatment components A_k simultaneously and without bias, which stratification may be unable to do even with data on all time-varying confounders.

Fine Point 20.2: Confounders on the Causal Pathway

Conditioning on a confounder L_1 affected by prior treatment can create selection bias even when L_1 is not on a causal pathway from treatment to outcome (no such pathway exists in Figures 20.5 and 20.6).
In Figure 20.7, by contrast, L_1 is a confounder for A_1 *and* lies on the causal pathway $A_0 \to L_1 \to Y$. Even if U_1 were not a common cause of L_1 and Y (no selection bias), the A-Y associations within strata of L_1 would estimate only the direct effect of A_0 not through L_1, not the overall effect of $\bar{A}$ on Y.
The common rule that variables on a causal pathway cannot be confounders is inaccurate for time-varying treatments: a confounder for later treatment A_1 can lie on a pathway from earlier treatment A_0 to Y. Whether adjusting for it induces bias depends on the method: stratification does; g-methods do not (Hernán and Robins 2020, Fine Point 20.2, p. 272).

4 20.4 Why Traditional Methods Cannot Be Fixed (pp. 273-274)

Could parametric **outcome regression** succeed where nonparametric stratification failed? The question matters most with high-dimensional data, where a simple stratified analysis is impossible.

Example 4.1 (Counting Static Strategies)**.**

- With two time points there are $2^2 = 4$ static strategies $\bar{a}$.
- With about 100 time points (not unusual in practice), there are about 2^{100} static strategies, far more than the sample size of any study, and even more once dynamic strategies are considered.

The number of combinations of the *data* is larger still, because several confounders L_k are measured at each time point.

4.1 Modeling Is Needed, but Does Not Help

Remark 4.1 (Many Strategies Require Modeling). As argued since Chapter 11, many possible strategies require **modeling**: a dose-response function for the effect of treatment history $\bar{a}$ on the mean outcome. For example, assume the effect is linear in **cumulative treatment**, so every strategy with exactly three months of treatment has the same effect, whenever those three months occur. The price is a new threat to validity: misspecification of the dose-response model.

⚠ Conventional Outcome Regression Does Not Remove the Bias

Paying the price of a dose-response model buys no protection if we read the treatment coefficients of an outcome regression as the effect while the model conditions on the time-varying confounder. That conventional use of regression **is** a stratification-based method, so it cannot remove the bias of stratification under treatment-confounder feedback (Definition 1.1). The g-formula (Chapter 21) can fit the same kind of outcome model, but it averages the model's predictions over the distribution of each time-varying confounder given past treatment and past confounders (in this chapter's two-time-point example, which has no L_0: L_1 given A_0) instead of reading the effect off the coefficients.

4.2 Example: A Cumulative-Treatment Regression

Example 4.2 (A Cumulative-Treatment Regression)**.** For data generated under Figure 20.5, define $\text{cum}(\bar{A}) = A_0 + A_1 \in \{0, 1, 2\}$. "Always treat" is $\text{cum}(\bar{a}) = 2$ and "never treat" is $\text{cum}(\bar{a}) = 0$; the target is $\mathbb{E}\left[Y^{\text{cum}(\bar{a})=2}\right] - \mathbb{E}\left[Y^{\text{cum}(\bar{a})=0}\right]$, whose true value is 0.
Fit the outcome regression model

$$\mathbb{E}\left[Y \mid \bar{A}, L_1\right] = \theta_0 + \theta_1 \,\text{cum}(\bar{A}) + \theta_2 L_1.$$

Within either level l of L_1,

$$\begin{aligned}&\mathbb{E}\left[Y \mid \text{cum}(\bar{A}) = 2, L_1 = l\right] - \mathbb{E}\left[Y \mid \text{cum}(\bar{A}) = 0, L_1 = l\right] \\ &\quad = (\theta_0 + 2\theta_1 + \theta_2 l) - (\theta_0 + 0 \cdot \theta_1 + \theta_2 l) = 2\theta_1.\end{aligned}$$

⚠ Do Not Read $2\theta_1$ as the Causal Effect

It is tempting to read $2\theta_1$ as the effect of "always treat" versus "never treat" within levels of L_1. But conditioning on L_1 generically induces an association between A_0 (a component of $\text{cum}(\bar{A})$) and Y, so, generically, $\theta_1 \neq 0$ even though the true effect is zero (Hernán and Robins 2020, 274). The bias comes from conditioning on the collider L_1, not from the form of the model or from the sample size: it would persist with unlimited data.

A similar argument applies to **matching**. G-methods are needed to adjust appropriately for time-varying confounders in the presence of treatment-confounder feedback (Hernán and Robins 2020, 274).

Example 4.3 (A Correctly Specified Outcome Model for the Trial Data)**.** The book states that θ_1 is non-zero "even if the true causal effect is zero and the regression model for $\mathbb{E}\left[Y \mid \bar{A}, L_1\right]$ is correct" (Hernán and Robins 2020, 274). Under Figure 20.5, the cumulative-treatment model itself generically cannot be the correct one. Y is independent of A_1 given (A_0, L_1), so a correct model for $\mathbb{E}[Y \mid A_0, A_1, L_1]$ has no A_1 term, whereas $\text{cum}(\bar{A})$ forces A_0 and A_1 to share the coefficient θ_1. The data in Table 1 illustrate this. Their cell means depend on (A_0, L_1) only, and they are fit exactly by

$$\mathbb{E}[Y \mid A_0, A_1, L_1] = 84 - 8A_0 + 0 \cdot A_1 - 32L_1,$$

which reproduces all four distinct cell means:

- $A_0 = 0, L_1 = 0$: 84;
- $A_0 = 0, L_1 = 1$: $84 - 32 = 52$;
- $A_0 = 1, L_1 = 0$: $84 - 8 = 76$;
- $A_0 = 1, L_1 = 1$: $84 - 8 - 32 = 44$.

The conclusion is unchanged: even this correctly specified model gives A_0 a non-zero coefficient (-8) although the true effect of A_0 is zero, because the model conditions on the collider L_1.

5 20.5 Adjusting for Past Treatment (pp. 274-275)

Remark 5.1 (Past Treatment Opens New Backdoor Paths). So far the diagrams had no arrow $A_0 \to A_1$. Now suppose doctors use past treatment history $\bar{A}_{k-1}$ when deciding on treatment A_k. Adding $A_0 \to A_1$ to Figures 20.3 and 20.4 gives the book's Figures 20.8 and 20.9.
Under treatment-confounder feedback (Definition 1.1), adjusting for L_1 alone no longer closes every backdoor path from A_1 to Y. L_1 is a collider on each of the following paths, so conditioning on it opens them:

$$A_1 \leftarrow A_0 \to L_1 \leftarrow U_1 \to Y \qquad \text{(Figure 20.8)},$$
$$A_1 \leftarrow A_0 \leftarrow W_0 \to L_1 \leftarrow U_1 \to Y \qquad \text{(Figure 20.9)}.$$

And whenever past treatment affects the outcome (Figure 20.10), A_0 is a confounder for the effect of A_1, feedback or not.

Remark 5.2 (Sequential Exchangeability Conditions on Treatment History). Sequential exchangeability at time k generally requires conditioning on the treatment history $\bar{A}_{k-1}$ as well as on the covariates; conditioning only on L is not enough. That is why every sequential exchangeability statement in Chapters 19 and 20 conditions on treatment history.

5.1 Past Treatment and Time-Fixed Treatments

Definition 5.1 (Short-Term Effect)**.** In a two-time-point treatment $\bar{A} = (A_0, A_1)$, the **short-term effect** of A_1 is the contrast

$$\mathbb{E}[Y^{a_1=1}] - \mathbb{E}[Y^{a_1=0}],$$

in which only A_1 is set by intervention and A_0 is not intervened on, so it keeps whatever value it would naturally take. That is, it is the effect of A_1 when A_1 is treated as a time-fixed treatment.

Suppose the target is the short-term effect of A_1 (Definition 5.1), treating A_1 as a time-fixed treatment.

⚠ Ignoring Past Treatment Biases the Short-Term Effect

Without adjustment for A_0:
- with treatment-confounder feedback, there is generically (under faithfulness) selection bias;
- if A_0 affects Y directly, there is generically confounding.

So $\mathbb{E}[Y \mid A_1 = 1, L_1] - \mathbb{E}[Y \mid A_1 = 0, L_1]$ need not be zero even if A_1 has no effect on anyone's outcome (Figures 20.8-20.10).

💡 Adjust for Treatment History or Restrict to New Users

In practice, this bias tends to show up when current users ($A_1 = 1$) of a time-fixed treatment are compared with nonusers ($A_1 = 0$). Either of two remedies avoids it:
- adjust for the prior treatment history;
- keep only individuals who share one particular treatment history.

New-user designs take the second route, keeping only people who have never used the treatment before. The restriction is a stand-in for adjustment: an analysis that already adjusts

correctly for past treatment gains nothing against this bias by also discarding prevalent users.

5.2 Mismeasured Past Treatment

⚠ Mismeasured Past Treatment

The need to adjust for past treatment matters when past treatment is **mismeasured**. As in Section 9.3[a], adjusting for a mismeasured confounder can leave bias in either direction.

[a] 09-measurement-bias.qmd#mismeasured-confounders-and-colliders-pp.-128-130

Example 5.1 (Self-Reported Treatment History)**.** Suppose HIV investigators lack medical records and ascertain prior treatment by questionnaire, so they observe a mismeasured A_0^* rather than A_0. Adding the arrow $A_0 \to A_0^*$ to Figures 20.8-20.10 shows the problem. A_0^* is only a noisy child of A_0, so conditioning on it leaves open the backdoor paths from A_1 to Y on which A_0 lies. Investigators would generically (under faithfulness) find an association between A_1 and Y even after adjusting for A_0^* and L_1, although A_1 has no effect on Y.

⚠ Measurement Error Can Bias Even Under the Null

With a time-varying treatment, errors in recording treatment **can bias the estimate when treatment has no effect**, even if the errors are independent and non-differential, although such errors are often assumed to be harmless under the null. Earlier treatment confounds later treatment whether or not it affects the outcome, and an error-prone record of earlier treatment removes that confounding only in part. When treatment does have an effect, the same partial adjustment can overstate it. Robins (1987) showed that random errors in measured treatment can push estimates away from the null (Hernán and Robins 2020, 275).

6 Summary

- **Treatment-confounder feedback** (Definition 1.1): a time-varying confounder L_k affects later treatment A_k and, for some earlier treatment A_j ($j < k$), either is affected by A_j or shares with A_j a cause outside the measured past of A_j, through a path that conditioning on that measured past does not block.
- In Table 1 the true effect of "always treat" vs. "never treat" is 0, yet the unadjusted analysis gives -13.3 and stratification on L_1 gives -8.
- Stratification on L_1 removes confounding for A_1 but opens the collider path $A_0 \to L_1 \leftarrow U_1 \to Y$, creating selection bias for A_0.
- Conventional outcome regression (reading the effect off treatment coefficients in a model that conditions on L_1) and matching are stratification-based too, so they are biased even when the regression model is correct.
- Sequential exchangeability generally requires conditioning on past treatment; mismeasured past treatment can bias estimates even under the null.
- G-methods (Chapter 21) handle treatment-confounder feedback correctly.

7 References

Hernán, Miguel A, and James M Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC. https://miguelhernan.org/whatifbook.

Robins, James M. 1986. "A New Approach to Causal Inference in Mortality Studies with a Sustained Exposure Period – Application to Control of the Healthy Worker Survivor Effect." *Mathematical Modelling* 7: 1393–512.

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